

Seed bankroll proposal: \$20,000 for the rest of the 2026 college football season

Prepared September 17, 2026. This is a private request to family for a **gift** that funds a betting bankroll. Nothing is owed back. It is not an investment, not a loan, and not a security. What follows is what the money would fund, what has been earned so far, and what the model says is likely to happen to it.

Every number below comes out of a script in this repository. Reproduce with:

```
python research/bankroll/scripts/mc_combined_totals.py --paths 100000
python research/bankroll/scripts/bankroll_config_sweep.py --paths 50000 --out research/bankroll/docs
```

1. The ask

item	value
amount	\$20,000
horizon	weeks 4–15 of the 2026 regular season, September 24 to December 12. Twelve weeks. Week 3 (this weekend) is not in the projection
what it funds	two totals strategies, both already running, at flat stakes
expected bets	~99: ~88 Greenline unders and ~11 over-zero overs
recommended stake	Greenline \$100 per bet (0.5%), over-zero \$200 per bet (1%)
what happens after	bankroll and profit stay in the operation for 2027, where the larger of the two edges does most of its work

2. The answer in one paragraph

Median outcome is a gain of \$590 to \$820 over twelve weeks. Roughly one season in three ends below \$20,000. No modeled path loses a quarter of the money, and none goes to zero. The range is bracketed because the main strategy's win rate rests on two defensible readings of the record, and neither can be ruled out yet. Everything below reports both.

Greenline prior	median	90% band	P(down)	P(–25%)	busts
pooled (141–109)	\$20,821 (+4.1%)	\$18,685 – \$22,945	26.3%	0.0%	0.0%
n49 (27–22)	\$20,594 (+3.0%)	\$17,818 – \$23,297	36.2%	0.0%	0.0%

3. What gets bet

Two strategies. One bankroll. Flat stakes off the starting \$20,000, no compounding.

Over-zero (floor-bias OVERs)

A model built in this repository. It finds games where the market total is pinned too low by the way books price a heavy favorite's opponent, and bets the OVER when the expected bias clears a threshold.

fact	value	source
walk-forward record, 2016–2025	151–83, 64.5% (95% CI 58.2–70.4%)	models/over_zero/docs/ROI_HITRATE.md
ROI at –110	+23.2% (CI +11.1 to +34.4%)	same
planning win rate	58.2% , the CI floor, because the threshold was picked on this data	MODEL_GUIDE.md
price rule	–120 or better	same
bets left in 2026	~ 11 . Volume is front-loaded into weeks 1–3, already played	mc-combined-totals-2026-09-17.md
2026 so far	17–8 through week 2	site/lib/weekly-results.json

This is the better-evidenced edge, and it is nearly spent for 2026. Its median contribution from here is **+\$133**. It matters for 2027, when it fires 30–50 times.

Greenline totals (PFF vendor flags, unders)

PFF publishes a projection per game and flags totals where it disagrees with the market. Flags are captured Wednesday, graded Monday. ~85% of flags are unders.

fact	value	source
2026 graded flags	27–22, 55.1% (CI 41–68%), one week	greenline-season-review-2026-09-16.md
2023–25 personal unders, mostly the same flags	114–87, 56.7% (CI 49.8–63.4%), 201 bets	bet-history-analysis-2023-2025.md
pooled prior	141–109, posterior mean 56.4%, P(losing) 10%	mc-combined-totals-2026-09-17.md
2026-only prior	27–22, posterior mean 55.0%, P(losing) 35%	same
historical rate bet	~13% of flags, ~ 7 a week , ~88 over 12 weeks	same
price	–110, break-even 52.38%	same

The 2023–25 record is the same signal in earlier seasons, bet by the same person. It is legitimate prior evidence. It is **not** independent confirmation of PFF, and this document never treats it as one. Those 201 unders also sat six points higher in total than the 2026 flags (median 58.5 vs 52.5), so the pooled 56.4% probably reads high. That is why the 2026-only prior is carried alongside it everywhere.

Conflict rule

Rare, but it has happened: week 2, Rice @ Notre Dame, over-zero said OVER 54.5 and Greenline flagged UNDER 55.5. Betting both pays juice twice for a hedge. **Rule: Greenline takes the game, over-zero skips it.** Greenline carries the volume.

Excluded

The spread model in research/spread/ is research, not a bet. Greenline spreads and moneylines went 21–28 and 21–25 in week 2 and are not funded.

4. How the projection works

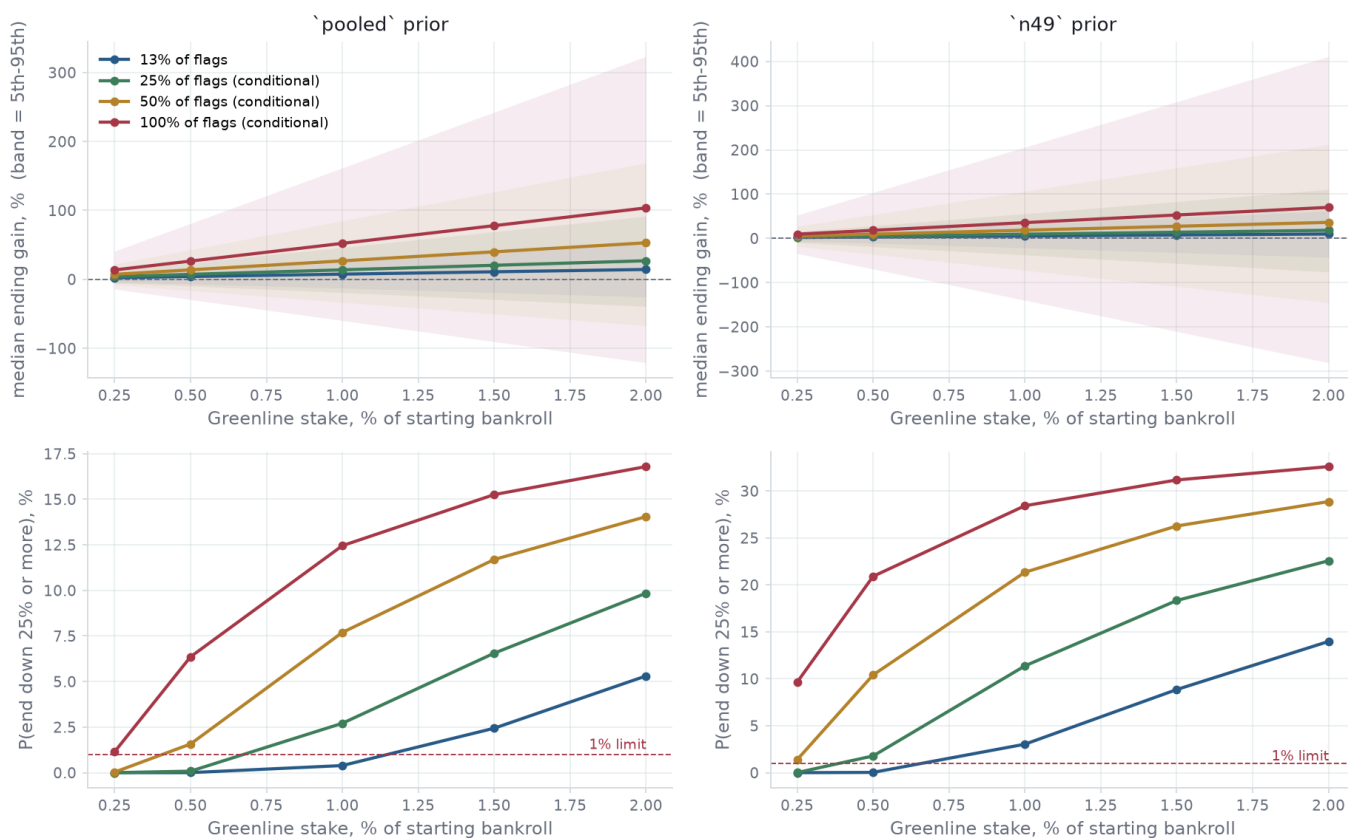
One simulated path is one whole remainder-of-season. 100,000 paths for the headline, 50,000 per cell of the sweep.

- **Win rates are never fixed.** Each path draws its own win rate from the Beta posterior of the graded record, so the spread of outcomes includes not knowing the true rate, not just luck.
- **Bets on the same Saturday are correlated** through one scoring-environment shock (Gaussian copula, $\rho = 0.10$ assumed, $\rho = 0.25$ as sensitivity). Because over-zero is all overs and Greenline is mostly unders, a high-scoring day helps one and hurts the other.
- **Volume follows the schedule.** Greenline flags every FBS-vs-FBS game (week 2: 49 games, 49 flags; week 3: 57 and 57), so the weekly flag count is the slate: 56–67 games a week through week 14, then 9 in championship week, ~680 flags in all. Bets are Poisson around 13% of that. Over-zero resamples its week-4+ history (8, 11, 9, 11, 14 bets in 2021–25).
- **Flat stakes off the starting bankroll**, no stop-loss. A path can go through zero and keep betting, so the bust rate is reported on every row.
- Pushes not modeled: zero realized on all 484 graded bets, all on half-point lines.

5. Choosing the stake: the sweep

Greenline stake from 0.25% to 2% of bankroll, coverage from the historical 13% up to every flag, both priors. Over-zero held at 1%. Full grid in [bankroll-config-sweep-2026-09-17.md](#).

Stake x coverage sweep, \$20,000, weeks 4-15 of 2026



Objective (a): largest median gain such that at most 1% of paths end down 25% and none go through zero, under **both** priors. **Objective (b):** downside ratio = median gain ÷ (median – 5th percentile). Shown as a column; higher is better.

Supported rows only (13% coverage, the population the record came from):

GL stake	median (pooled / n49)	5th pct (pooled / n49)	P(–25%) (pooled / n49)	ratio (pooled / n49)	passes (a)
0.25%	\$20,480 / \$20,367	\$19,144 / \$18,750	0.0% / 0.0%	0.36 / 0.23	yes
0.50%	\$20,821 / \$20,594	\$18,685 / \$17,818	0.0% / 0.0%	0.38 / 0.21	yes
1.00%	\$21,502 / \$21,052	\$17,479 / \$15,736	0.4% / 3.0%	0.37 / 0.20	pooled only
1.50%	\$22,182 / \$21,503	\$16,197 / \$13,591	2.4% / 8.9%	0.36 / 0.19	no
2.00%	\$22,864 / \$21,952	\$14,879 / \$11,421	5.3% / 14.0%	0.36 / 0.19	no

Three things the grid says:

- 0.5% is the largest stake that passes under the conservative prior.** 1% passes only if the pooled prior is right. Recommended: **\$100 per Greenline bet**, with \$200 as the upgrade once the 2026 flags alone reach ~250 graded (four more weeks).
- Stake does not change the downside ratio.** It runs 0.36–0.38 under pooled and 0.19–0.23 under n49 at every stake. Staking more buys a bigger median and a bigger 5th-percentile loss in the same proportion. There is no free stake.
- Coverage is what improves the ratio** (0.36 → 0.49 under pooled at 0.25%), and coverage is exactly the assumption the record does not support. Betting more of the flags is the lever, and it is unpriced until the ledger says which flags get taken. With volume following the slate, every row above 13% coverage now fails the risk limit under the conservative prior except 0.25% at 25%, which earns less than the recommended row. So the conditional rows are shown for the tradeoff and none is a candidate.

6. Risk, stated plainly

At the recommended \$100 / \$200 stakes:

measure	pooled	n49
P(season ends below \$20,000)	26.3%	36.2%
P(ends below \$15,000)	0.0%	0.0%
P(passes through \$0)	0.0%	0.0%
5th-percentile ending bankroll	\$18,685	\$17,818
worst single week, median	–\$461	–\$509
total staked over 12 weeks	~\$10,900	~\$10,900

The 5th percentile means one season in twenty ends worse than about –\$1,300 to –\$2,200. The 30-ish percent chance of a losing season is mostly parameter uncertainty: the true Greenline win rate has a 10–35% chance of being below break-even, and the season is too short to average that away.

Correlation only bites the tail. Median is flat across p ; the 5th percentile moves by a few hundred dollars. p is assumed, not measured.

7. What this does not support

- **Greenline as independently validated.** $n=49$ in 2026, CI 41–68%. The pooled prior is 80% personal history of the same signal.
- **The pooled prior transferring in full.** The 201 unders sit ~6 points higher in total than the 2026 flags. Pooling probably overstates; by how much is unknowable from what exists.
- **Any coverage above 13%.** Every conditional row assumes the picked-flag win rate applies to flags that were passed on.
- **A reproducible selection rule.** "Bet ~13% of the week's flags" is a volume assumption. No script picks which six. The historical picks were hand-filtered and price-shopped.
- **A 2027 projection.** Not modeled. Over-zero at full-season volume and a full graded Greenline season are what it needs, and neither exists yet.
- **Compounded returns.** Simultaneous Saturday kickoffs make them unachievable; every figure here is flat-staked.

8. What the money does each week

day	step	command
Wednesday	capture Greenline flags	<code>scripts/pull_pff_scoreboard.py --greenline</code>
Wednesday	seed the bet ledger	<code>research/bankroll/scripts/greenline_bet_log.py --seed</code>
Thursday–Saturday	bet ~7 unders at –110 or better, over-zero board at –120 or better	<code>models/over_zero site</code>
Monday	grade flags, mark which were bet	<code>grade_greenline.py ; greenline_bet_log.py --mark</code>
Monday	rerun the projection with the new record	<code>mc_combined_totals.py</code>

Marking which flags get bet is the one step that is not automated and the one that resolves the biggest open question. Four more graded weeks gets the 2026 flags to $n \approx 250$ on their own, at which point the prior stops doing the work and the stake decision in \$5 gets revisited.

Data and dates

over-zero: 234 walk-forward bets 2016–2025, `models/over_zero/docs/backtest_bets.csv`. Greenline: 49 graded flags, 2026 week 2, `data/ingest/pff_scoreboard/greenline_graded.csv`; 201 personal unders 2023-08 to 2025-12, `data/ingest/bet_history/history.csv`. Projection span weeks 4–15, 2026. Seeds 20260917. Sweep: 50,000 paths per cell, 40 cells; headline table §2 from the sweep's 0.5% rows. Greenline volume per week from `core.fact_game` FBS-vs-FBS counts (weeks 14–15 from 2025), queried 2026-09-17.

Related: [mc-combined-totals-2026-09-17.md](#), [under-selection-profile-2026-09-17.md](#), [bankroll-config-sweep-2026-09-17.md](#).